

INTERPRETABLE DESTINATION-AWARE SYNTHESIZER MODULATION RECOVERY

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ABSTRACT

Synthesizer programming is a challenging task, particularly in configuring modulation: the time-varying control of parameters such as pitch, filter cutoff, and oscillator level by modulators such as Low-Frequency Oscillators (LFOs). Prior work has shown that modulation can be reconstructed from the clean audio through curve matching, yet the recovered parameters are often untransferable to modern synthesizers. We present an interpretable destination-aware modulation and waveform recovery pipeline that mirrors how musicians often recreate sounds on a synthesizer. First, the system identifies the modulated destinations; then it recovers the LFO shape associated with each destination, and estimates the oscillator waveform to match the source timbre. The training is supported by a differentiable synthesizer that incorporates a noise oscillator, and more modulation options than previous work. We carefully train the models for optimal perceptual quality, with extensive exploration of perceptual losses and adversarial training. Through objective and subjective evaluation, we show that predicting the destinations correctly is essential, our model excels in the modulation focused inverse synthesis tasks, and our Gammatone loss and CQT discriminator significantly outperform the traditionally-used MSS loss in improving perceptual quality. We provide audio samples from the subjective test¹.

Index Terms— Inverse Synthesis, Modulation Discovery, Differentiable DSP, Classification

1. INTRODUCTION

Synthesizer programming is the process of configuring a synthesizer’s parameters to get a desired audio output. When a music producer wants to use a sound they hear, they need to recreate the patch for the synthesizer, which can be challenging. Modern synthesizers such as Serum² and Vital³ are very expressive, exposing hundreds of parameters and modulation destinations, with many contemporary presets relying heavily on dynamic control signals such as LFOs. As a result, recreating a sound has become increasingly difficult, especially because synthesizer programming is fundamentally a many-to-one problem: multiple distinct parameter configurations can produce perceptually similar audio.

Recent advances in differentiable signal processing have significantly expanded the possibility of inverse synthesis programming. Early differentiable synthesizers [1–5] implemented classical synthesis techniques [6, 7] as differentiable modules, enabling end-to-end optimization of synthesis parameters directly from audio.

Work done as part of an internship in Dolby. Thanks to Daniel Arteaga for providing insightful opinions during the project. LLMs were only used to assist with coding and grammar checking.

¹<https://dolby.box.com/v/dest-aware-MUSHRA>

²<https://xferrecords.com/products/serum-2>

³<https://vital.audio/>

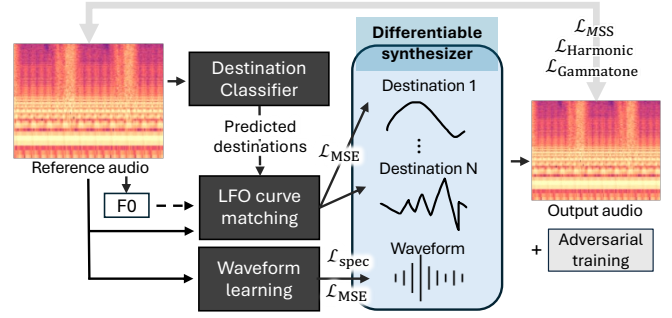


Fig. 1. An interpretable inverse synthesis system featuring three major modules, which are connected by the differentiable synthesizer and trained with advanced techniques. Dotted lines indicate conditioning.

While demonstrating capability in modeling real-world and synthetic sounds, these approaches faced limitations when applied to synthesis inversion for modern synthesizers. To address this gap, subsequent work introduced more complex differentiable synthesizers incorporating modern modules [8–10] and a broader range of controllable parameters [11, 12]. These systems support the inversion of multiple synthesizer parameters, predominantly in an end-to-end fashion via parameter regression losses [8, 12–14], while others have turned to generative approaches [15, 16] that trade interpretability for greater generative power.

To further model the temporal expressiveness of sound, a line of work focuses on emulating the time-varying features of audio [17, 18]. In particular, ModSynth [19] targets arbitrary modulation control, supporting three modulation types and providing an algorithm to extract arbitrary modulation curves (LFOs) built upon LFO-net [20], evaluated across white-, gray-, and black-box scenarios. Despite these advances, critical gaps remain. First, existing systems perform modulation extraction only on fixed destinations and clean audio, limiting their generalizability to practical use cases involving arbitrary destinations and more intricate sounds. Second, no existing method has attempted a holistic sound matching approach that simultaneously addresses LFO recovery across different destinations and waveform matching with perceptual losses.

In this work, we aim at an interpretable inverse synthesis system that combines destination detection, modulation extraction, and waveform learning. As illustrated in Fig 1, our approach mirrors the process a musician would follow when attempting to recreate a sound on a synthesizer: first identifying which modulation destinations are active, then characterizing the shape of the modulation applied to each destination, and finally matching the underlying oscillator waveform to reproduce the source timbre. Waveform recovery is a non-trivial step in this pipeline, as the modulations applied to a sound continuously alter its spectral shape over time, making it difficult to directly infer the underlying oscillator waveform from

the audio alone. We also attempt to tackle the inherent many-to-one issue by an extensive exploration of perceptual losses and including phase-invariant loss for waveform learning.

2. METHODS

2.1. Custom Synthesizer Design and Setup

The synthesizer is a fully differentiable subtractive architecture supporting gradient flow from the audio output back through all the components, enabling end-to-end training against audio-domain losses. The signal path is as follows: oscillator and noise sources are amplitude-scaled and mixed, passed through a highpass then low-pass biquad filter in series, multiplied by an envelope, decimated, and finally RMS-normalized with a soft tanh limiter. Biquad filters are applied using `torchlpc`⁴ [21], which provides a parallelized recursive IIR filter implementation.

Active Destinations. The modulation matrix routes LFO signals to five synthesis destinations, modulating parameters within the following ranges: 1) Oscillator level (-40 to +6 dBFS); 2) Oscillator coarse pitch (± 12 semitones); 3) Noise level (-40 to +6 dBFS); 4) Highpass cutoff (0.2–8kHz); 5) Lowpass cutoff (0.2–18kHz).

Oscillator/LFO curves. The oscillator produces one-cycle waveform. The LFO curves are loops with certain period. Both waveform and LFO are sampled at 22kHz. All of the LFOs are absolute, except for pitch, which is relative to the base pitch.

2.2. LFO and Waveform Recovery Pipeline

Step 1: Destination Classification. Given an input audio signal, the first step in this pipeline is to identify the destinations that are actively modulated. We train a multi-label classification model for this task, as it is common for more than one destination to be modulated simultaneously in a sound. The backbone model is expected to: 1) capture the necessary acoustic features of the input signal, and 2) model the signal’s temporal context over timescales of up to several seconds in order to detect modulation patterns over time. We choose an AST-style [22] classifier for this task. AST is a purely attention-based model originally designed for audio classification. Its self-attention layers provide the ability to capture long-range global context. We adapt and train the AST model from scratch to have a smaller hidden dimension (128 vs. 768) and fewer number of layers (4 vs. 12). The number of parameters in this smaller model is around 800k, while achieving similar performance on our task. The destination classification model is trained independently from the subsequent regression models, supervised by a binary cross-entropy loss.

Step 2: Destination-conditioned LFO Regression. We activate zero to two simultaneous modulation destinations per sample during training. The regression model must isolate the contribution of the queried destination while suppressing interference from the concurrent modulation. We adopt the SpectralCNN2D architecture from the previous work [19] and condition it on the target destination by embedding a destination representation directly into the input spectrogram prior to the convolutional layers. This allows a single model to handle all destinations jointly. In addition to destination conditioning, we also use f_0 conditioning, in order to give context on the base frequency, specifically for relative pitch modulation. The f_0 conditioning is added to the spectrogram in the same way as the destination conditioning, but also added to the output projections for a stronger conditioning. We directly apply the MSE loss for the LFO curves matching, and process the entire spectrogram as one batch.

We found that chunking the input into small windows and reducing input length doesn’t improve the modeling accuracy.

As a post-processing step, we apply a multi-stage cleaning pipeline that first smooths the raw LFO signal using a sliding window average, then detects structurally meaningful keypoints by identifying peaks, troughs, and inflection points via first and second derivative analysis, and finally reconstructs a clean fitted curve through those keypoints using either linear interpolation or cubic splines depending on how much the segment deviates from linearity.

Step 3: Waveform Modeling. In addition to LFO recovery, we model the base oscillator waveform to capture the timbral characteristics of the synthesizer. Because the oscillators only produce short one-cycle waveforms, we treat the time-domain samples of the waveform as parameters, as in previous wavetable learning methods [23, 24], and model the parameters through shape-matching. We use the same architecture as the LFO recovery model, but without any destination or f_0 conditioning. We first experimented with a sub-million small model which fails to reliably recover the waveform, with a notable degradation in perceptual metrics, specifically the high frequency harmonic content. Increasing model capacity resolves this, suggesting that waveform recovery requires sufficient representational capacity to disentangle timbral features from the modulation activity present in the input spectrogram. We also incorporate phase-invariant losses for the waveform by taking the MSE distance between the magnitudes of the FFTs ($\mathcal{L}_{\text{spec}}$). This ensures we do not penalize different phases, as their sounds are similar.

Pitch Complications. Pitch modulation introduces two distinct challenges that require targeted treatment. First, when pitch is actively modulated, the oscillator waveform is continuously stretched and compressed in the time domain, making it harder to recover the starting waveform. Second, the recovered LFO curves for pitch destinations tend to match the correct shape but are vertically offset from the ground truth shape. To address the vertical offset, we shift the recovered pitch LFO curves to align their mean with the f_0 derived from CREPE [25], leaving the shape of the curve intact. Increasing the waveform model size also fixed the issue with the waveform recovery. This results in better pitch matching in the resynthesized audio without having to retrain a separate or larger model specialized for pitch modulation.

2.3. Perceptual Quality Improvement

We add perceptual losses to co-train LFO and waveform regression to improve the perceptual quality, and tackle the many-to-one issue. We first applied multi-scale spectral (MSS) loss following the previous work [1, 2, 19], and observed reduced quality, particularly a tendency of pitch mismatching which has been studied before [26]. Thus, we further explored more perceptual treatment for this task, including harmonic loss, multiscale mel-spectrogram loss and Gammatone loss, and adversarial training.

Harmonic loss penalizes differences in the relative harmonic structure. For each signal, a differentiable f_0 is first estimated, upon which it constructs a harmonic vector. The loss is the mean L1 distance between the target and synthesized harmonic vectors.

Multiscale mel-spectrogram loss is adapted from MSS, computed across multiple mel scales and STFT sizes to capture perceptual differences at varying levels of time-frequency resolution.

Gammatone loss computes reconstruction error in the domain of a fixed, ERB-spaced Gammatone filterbank [27], approximating the frequency selectivity of the cochlea. A bank of $C = 64$ FIR filters is constructed from IIR Gammatone impulse responses with center frequencies distributed on the ERB scale between 50 Hz and Nyquist.

⁴<https://github.com/DiffAPF/torchlpc>

Each filter output is half-wave rectified and collapsed to a per-band RMS envelope over non-overlapping frames, yielding a representation $\mathbf{E} \in \mathbb{R}^{B \times C \times T}$. The loss combines a linear and log-domain L1 term over the envelope, where $\lambda = 1.0$ and $\epsilon = 10^{-5}$.

$$\mathcal{L}_\gamma = \left\| \mathbf{E} - \hat{\mathbf{E}} \right\|_1 + \lambda \left\| \log(\mathbf{E} + \epsilon) - \log(\hat{\mathbf{E}} + \epsilon) \right\|_1$$

Adversarial Training. We augment the reconstruction losses with an adversarial training objectives. We evaluate several discriminator architectures commonly used in neural audio synthesis: the Multi-Period Discriminator (MPD), Multi-Scale Discriminator (MSD), and Multi-Resolution Discriminator (MRD), and a Constant-Q Transform (CQT) discriminator [28, 29], which operates on a logarithmically-spaced time-frequency representation better suited to musical signals.

3. EXPERIMENT

3.1. Dataset and Training

We generate synthetic datasets following the spline-based LFO shape procedure from Mitcheltree et al. [19]. Each sample is synthesized using one of four canonical waveform shapes (sine, square, triangle, sawtooth) or a waveform drawn from the Serum *analog* or *spectral* wavetable banks. Task difficulty is controlled by varying synthesizer parameters specific to each experimental condition, as described in the following subsections. The training data consists of samples with 1 second loops. A single loop is given to the model. We experimented with varying loop lengths and found the model can extract the loop of different length well. The audio sample rate is 22.05 kHz. We use two main datasets: the 0-2 active destination dataset (2Dest) and the 0-5 active destination dataset (5Dest). 2Dest is used for both training and testing, and 5Dest is used for training the classifier as well as direct testing using models trained on 2Dest.

We prepare both synthetic and Serum 2 data for testing. Serum 2 dataset is generated [30] through Pedalboard [31] with the following configuration: one waveform oscillator, noise oscillator, filters, and LFO shapes generated in the same way as the synthetic datasets. The Serum 2 data under the same setup sound more refined compared to the synthetic datasets due to some unknown programming in the software; We treat it as a gray-box dataset.

The classifier and LFO/waveform models were trained independently. First, we train the destination classification model on 5Dest. Next, we train the LFO and waveform models concurrently on 2Dest using the MSE losses and optionally perceptual losses, while being conditioned on the ground truth f_0 and active destinations. During inference, f_0 is estimated using CREPE [25] and destinations are predicted using the trained classifier before being used as conditioning for the LFO/waveform models.

All models were trained using the AdamW optimizer with a learning rate of $1e-4$ on 1 A100 until convergence. A batch size of 64 was used for all models, except those trained with adversarial losses, which used a batch size of 32. For the loss weights, we found that $W_{\text{LFO}} = 1, W_{\text{Waveform}} = 0.1$ works the best for the non-perceptual losses. For the perceptual losses, $W_{\text{Harmonic}} = W_{\text{Gammatone}} = W_{\text{Mel-Spectrogram}} = 1.0$ and $W_{\text{MSS}} = 0.5$.

3.2. Baseline and Evaluation

We designed a baseline and an ablation besides the losses variants to compare with our model. The baseline has a competitive setup, in which a separate model is trained for each destination, akin to

previous works [19, 20], thus no destination classifier is technically needed. All other design choices, including f_0 conditioning, remain the same with the proposed model. To make this baseline a fair comparison to our large models, we scaled the model size down so that the total number of parameters of every destination-specific models are similar to the single proposed model plus the classifier. We also trained it so that if a destination was not active, the target of a straight horizontal line at the correct value was given. This was done to prevent odd modulation curves when no modulation existed.

To test the function of the classifier itself, we design a separate ablation model, labeled as No-Conditioning (NC). The NC model keeps the same setup as the main model, but having both classification and f_0 conditioning removed, such that the LFO regression model has no information about which destination it is modeling. Both the baseline and NC model are trained with MSE losses only.

Objective evaluation. Following previous work [19], we use MSS, MFCC-L1 and FAD-Encdec-24k⁵ as metrics to evaluate the synthesis quality of our models, from the perspective of spectral content reconstruction, human auditory perception, and perceptual distribution similarity, comparing all the variants with the baseline.

Subjective evaluation. We designed a MUSHRA-style listening study, and modified the anchors from the standards catering to this specific task. For anchor 1, we used the same waveform as the ground truth, but no modulation on any destination. For anchor 2, we change the LFO shapes to be sine waves, for the same active destinations as the ground truth. We include five synthetic samples randomly selected from the test set with 1-5 active destination as well as 5 Serum 2 samples generated manually. We made sure the active destinations appeared in the test are balanced. Based on the objective scores and authors' listening, the best three models (MSE losses only, MSE losses+Gammatone, MSE losses+Disc) and the baseline are included in the test. 12 listening experts conducted the test.

4. RESULTS

4.1. Class Conditioning Analysis

Both the objective evaluation in Tab.1 and the MUSHRA scores in Fig.2 show that the baseline performs worse than all variants of our proposed model, demonstrating the importance of both classification and f_0 conditioning. The baseline models were specifically trained to avoid cross-destination interference; however, we still observed that they occasionally confuse LFOs from other destinations with the current target destination, or produce non-zero curves when the destination is inactive, both of which contribute to the degraded quality. We attribute this to the fact that the baseline must simultaneously perform destination identification and curve matching. This design is less interpretable and makes the model more data-hungry and harder to optimize effectively.

Interestingly, the NC model showed decent performance and outperformed baseline, which was unexpected, because the NC is not designed with the capability to correctly model every LFO curve. This result demonstrates that predicting correct destinations with a less accurate curve has more performance gain than better curve prediction with potential interference, highlighting the importance of the classifier.

4.2. Perceptual Loss Analysis

From Table 1 we find that the MSE + Gammatone setup performs best on MSS and MFCC-L1, followed very closely by the MSE-

⁵<https://github.com/microsoft/fadtk>

Table 1. Loss Function Configurations and Evaluation Metrics on Synthetic and Serum Data. The best metrics are bolded, and the second best are underlined.

Configuration	Synthetic 2Dest			Synthetic 5Dest			Serum 2 5Dest		
	MSS ↓	MFCC-L1 ↓	FAD ↓	MSS ↓	MFCC-L1 ↓	FAD ↓	MSS ↓	MFCC-L1 ↓	FAD ↓
MSE-only	0.487 ± 0.330	0.561 ± 0.348	0.898	0.973 ± 0.757	0.970 ± 0.635	0.899	2.445 ± 1.720	2.341 ± 0.685	49.386
MSE + MSS	0.892 ± 0.505	0.822 ± 0.369	6.604	1.261 ± 0.696	1.131 ± 0.546	7.118	2.529 ± 1.693	2.392 ± 0.613	73.872
MSE + Gammatone	0.469 ± 0.333	0.532 ± 0.349	1.865	0.961 ± 0.766	0.952 ± 0.648	1.865	2.443 ± 1.593	<u>2.350</u> ± 0.696	43.957
MSE + Harmonic	0.743 ± 0.356	0.798 ± 0.354	1.747	1.181 ± 0.703	1.169 ± 0.605	1.722	2.461 ± 1.696	2.400 ± 0.639	61.960
MSE + Mel-Spectrogram	1.080 ± 0.552	0.953 ± 0.385	13.272	1.413 ± 0.687	1.235 ± 0.531	14.360	2.570 ± 1.953	2.356 ± 0.680	91.314
MSE + Disc. (CQT)	0.520 ± 0.326	0.605 ± 0.347	0.591	1.000 ± 0.751	1.007 ± 0.631	0.591	2.445 ± 1.720	2.383 ± 0.750	<u>45.977</u>
MSE + Disc. (No CQT)	0.528 ± 0.330	0.613 ± 0.347	<u>0.775</u>	1.013 ± 0.758	1.017 ± 0.633	<u>0.700</u>	2.440 ± 1.553	2.374 ± 0.650	46.915
MSE + Gammatone + CQT	0.796 ± 0.442	0.790 ± 0.374	4.934	1.185 ± 0.674	1.124 ± 0.564	4.200	2.522 ± 1.863	2.439 ± 0.630	75.467
MSE + ALL	1.116 ± 0.740	1.150 ± 0.634	8.683	1.399 ± 0.645	1.284 ± 0.532	9.069	2.695 ± 2.149	2.405 ± 0.649	85.865
No Conditioning (NC)	1.034 ± 0.551	1.097 ± 0.571	2.554	1.466 ± 0.708	1.507 ± 0.725	4.558	2.705 ± 1.608	2.577 ± 0.666	59.371
Baseline (One Model Per Destination)	1.812 ± 0.728	1.901 ± 0.814	1.697	2.038 ± 0.717	2.106 ± 0.760	2.438	2.940 ± 0.941	2.792 ± 0.649	51.344

Table 2. Model architecture configurations and properties. Each model is trained with MSE + Gammatone losses. The models are evaluated on 2Dest. Scaling from *Small* to *Large* yields a substantial improvement. The waveform model reflects the same size as the LFO models (e.g. the small setup uses a small model for both LFO and waveforms). The baseline uses the same waveform model as the large setup.

Model	Params	Channels	Hidden	MSS	MFCC-L1
Small	0.8M	64×5	64	1.112 ± 0.537	1.030 ± 0.388
Medium	2.7M	64, 64, 128, 128, 128	128	1.052 ± 0.537	0.956 ± 0.387
Large	7.6M	64, 128, 128, 256, 256	256	0.469 ± 0.333	0.532 ± 0.349
Baseline	1.6M × 5	64, 64, 64, 64, 128	128	1.812 ± 0.728	1.901 ± 0.814

only setup, where no perceptual loss is used. The model trained with all discriminators, including CQT achieves competitive scores, especially on the FAD score, which we attribute to discriminators’ strength in modeling the data distribution. It is particularly interesting that, in the subjective test, the MSE losses have an edge overall. However, as it outperforms the other two perceptual loss configurations more evidently in the synthetic test set, the Gammatone loss is rated as the best system in the *Serum 2* test set. Since ground truth active destinations are used during training, we alleviate the many-to-one problem in the training setup, allowing the direct MSE waveform losses to be highly influential while the perceptual losses help finetune the output. In the gray-box setup with *Serum 2*, however, perceptual losses demonstrate their benefit.

MSS and mel-spectrogram losses suffer from the same failure modes, where high-frequency content dominates the loss and the resulting gradient is poorly correlated with perceptual distance between synthesizer patches. Harmonic loss can be very noisy due to samples with pitch modulation and incorrect f_0 predictions, particularly for samples containing noise or complex harmonics. Gammatone loss is the only perceptual loss to improve the metrics; it specifically models audio perception according to the human auditory system, which may explain its benefits over other perceptual losses.

Adding the CQT discriminator shows clear improvement over only using MPD, MSD, and MRD, which were designed for speech synthesis, not musical audio. The CQT’s log-frequency axis with constant relative bandwidth means harmonic relationships appear as consistent patterns regardless of fundamental frequency, making it a more natural fit for discriminating synthesizer sounds and providing stronger gradient signal during training.

We found in our experiments that training with perceptual loss alone does not work, most likely because the model has no direct

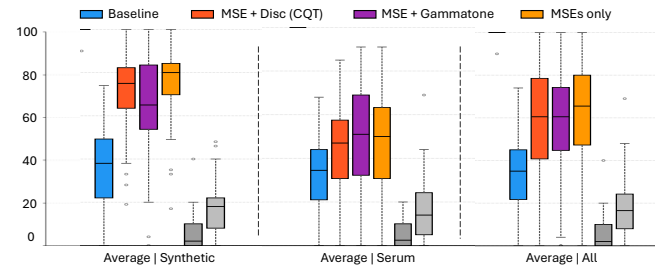


Fig. 2. MUSHRA results. Gray boxes refer to the anchors, and the hidden reference are all located at 100, except for one outlier that is above 90.

understanding of which synthesizer parameters affect which aspects of the sound, making it difficult to backpropagate meaningful gradients through the synthesis process. Introducing an auxiliary LFO or waveform loss appears necessary to ground the optimization in more direct parameter-level supervision. It is also worth noting that increasing the number of active modulation destinations within the synthesizer could potentially make perceptual loss far more useful, as a more expressive synthesis space would give the model greater flexibility to respond to perceptual feedback.

5. DISCUSSION

In this work, we presented a system that accurately identifies modulation destination, and reconstructs the LFO and waveform shapes of single oscillator audio, demonstrating that synthesizer reverse engineering is tractable even under noisy conditions. Moreover, the system delivers greatly improved synthesis quality with better perceptual treatment. However, the gap between our synthesizer model and professional synthesizers such as *Serum 2* highlights clear directions for future work. The two main limitations of our current system are the restricted set of filter types and the absence of wavetable oscillators. Similarly, we do not model effects such as reverb, distortion, or flanger, as time-varying effect parameters introduce a level of complexity we consider out of scope for this work. Currently, the model and synthesizer capacity does not match modern synthesizers, but the architecture allows for easy extension to additional modulation destinations. Future work will explore multiple oscillators, custom filter types, and effect modulation, moving toward a system capable of reverse engineering the full complexity of professional synthesizer patches.

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